

Principal Platform Software Engineer

Nashville, TN, United States

HOT JOB

- Job Identification340048
- Job CategoryProduct and Research
- Posting Date09/17/2026, 09:47 AM
- RoleIndividual Contributor
- Job TypeRegular Employee
- Does this position require a security clearance?No
- Years6 to 10+ years
- Applicants are required to read, write, and speak the following languagesEnglish

Job Description

The Oracle Cloud Infrastructure (OCI) team offers the opportunity to build and operate massive-scale, integrated cloud services in a broadly distributed, multi-tenant cloud environment. OCI builds cloud products for customers who are solving some of the world's largest technical and business challenges.

OCI Container Instances (CI) is OCI's managed serverless container offering. CI enables customers to deploy, operate and scale on OCI without the need for container orchestration, through seamless integration with OCI Compute, Networking, Storage, Identity, Security, Observability, and Developer services.

We are seeking a Principal Platform Software Engineer (IC4) who enjoys building reliable software systems and solving challenging engineering problems at the intersection of Linux, virtualization and cloud computing. This role is ideal for engineers with strong software development and system administration fundamentals.

As part of the CI team, you will design, develop, test, deploy and operate software that provisions, upgrades and manages containers across OCI regions. You will work on Linux virtualization, hypervisor development and imaging for containerized cloud workloads. The role requires strong expertise in QEMU/KVM, container runtimes, networking, storage, systemd, security and OS image creation. Engineers on this team have end-to-end ownership—from design and implementation through production operations—and contribute directly to the reliability, scalability, and customer experience of one of OCI's foundational cloud-native services.

Required Qualifications:

- BS/MS in Computer Science, Engineering, or equivalent practical experience.
- 6+ years of experience building and operating production software systems, preferably distributed systems or cloud services.
- Strong system administration experience in **Linux**, including **bash/shell scripting**.
- Strong programming experience in **Java** or **Go**.
- Strong fundamentals in data structures, algorithms, operating systems, networking, and distributed systems.

- Experience working with containers, Docker, Kubernetes, or related cloud-native technologies.
- Experience with CI/CD systems, automated testing, and modern software development practices.
- Experience using AI-assisted software development tools to improve engineering productivity across coding, testing, debugging, and documentation while maintaining high standards for code quality, security, and engineering judgment.
- Experience using Terraform for Infrastructure as Code
- Strong debugging, troubleshooting, and problem-solving skills.
- Excellent communication skills with a strong sense of ownership and the ability to collaborate effectively across teams.
- Experience operating highly available production services, including on-call rotations, incident response, and driving operational excellence.

Preferred Qualifications:

- Experience with QEMU/KVM: launching and managing VMs, working with virtual devices like virtio-net and virtio-iscsi, performance profiling, troubleshooting.
- Experience with container technologies like Docker or OCI containers and container runtimes such as crio, containerd/runc.
- Strong knowledge of Linux internals: namespaces, cgroups (v1 & v2), networking, and storage subsystems
- Hands-on experience with systemd: writing and managing unit files, controlling resource allocation, and understanding service lifecycles
- Ability to set up Linux bridges, veth pairs, and configure custom network namespaces and routing
- Experience with block storage: LUKS encryption, LVM logical volumes, and virtual disk provisioning (e.g., virtio-iscsi)
- Familiarity with security best practices for virtual machines and containers, including disk encryption and secure bootstrapping
- Experience in debugging with tools like strace, tcpdump, or wireshark, and analyzing logs. Ability to analyze core dumps with gdb would be a plus.

Responsibilities

As a Principal Software Platform Engineer on the Container Instances (CI) team, you will:

- Contribute to the Container Instances Data Plane team, designing, building, and maintaining production-grade Linux hypervisor and guest operating system images used to run containerized workloads.
- Develop and maintain automated processes for creating, customizing, validating, and publishing Linux images in formats such as qcow2.
- Customize Linux operating systems at the kernel, systemd, networking, storage, security and package-management layers to meet platform requirements.
- Build and maintain RPM packages, software repositories, boot configurations and image composition workflows using DNF/YUM and, where applicable, OSTree-based technologies.
- Develop and maintain hypervisor and lifecycle-management components in Java, Go and shell scripting.

- Design and optimize virtualization solutions using QEMU/KVM and paravirtualized devices, including virtio-based networking and storage.
- Profile and improve the performance, reliability, boot time, resource utilization and scalability of virtualized Linux environments.
- Troubleshoot complex issues across the Linux kernel, virtualization stack, container runtimes, networking, storage and distributed cloud infrastructure.
- Design and execute functional, integration, performance, regression and security testing for Linux images and virtualization components.
- Build and enhance CI/CD pipelines that automate image creation, package integration, security validation, testing, release qualification and deployment.
- Monitor and remediate operating system vulnerabilities by integrating security patches, kernel updates, package updates, encryption and secure-bootstrapping practices into the image lifecycle.
- Integrate the data plane with OCI services and APIs for networking, storage, identity, telemetry and infrastructure lifecycle management.
- Automate infrastructure provisioning and validation using Terraform and other Infrastructure as Code practices.
- Coordinate the safe rollout of new Linux image releases across hundreds of hypervisors in multiple global regions, including staged deployment, compatibility validation, monitoring and rollback planning.
- Produce clear technical documentation, design proposals, operational procedures, and troubleshooting guides, and communicate architectural decisions and technical tradeoffs to engineering partners.